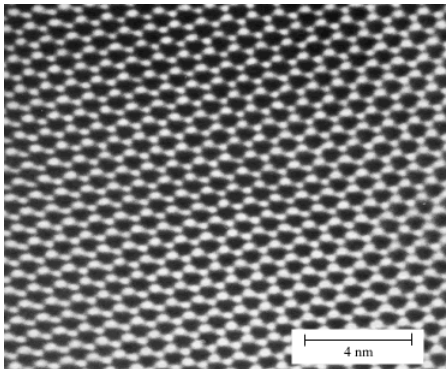


1.7.4: Introduction

Crystalline materials are characterised by a regular atomic structure that repeats itself in all three dimensions. In other words the structure displays **translational symmetry**. Translational symmetry is illustrated in this image of the crystal structure of the mineral cordierite ($\text{Mg}_2\text{Al}_4\text{Si}_5\text{O}_{18}$) taken with a high resolution transmission electron microscope. The image is a projection through a very thin slice (~ 200 Å thick) of the atomic distribution. Black spots correspond to hollow channels through the structure and white spots correspond to regions of high electronic density, arranged around the channels in 6-fold rings (Scale: the distance between the black spots is ~ 9.7 Å). The structure consists of a simple group of atoms that repeats itself periodically in space. This periodicity can be revealed using the concept of a **lattice**.



A. Putnis, *Introduction to Mineral Sciences*, Cambridge University Press, 1992 - frontispiece

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